

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

How to Configure Linkage Tracking For Thermal PTZ Cameras

SUPPORTED PRODUCT LIST

1. 41x7 series: W Series 8-inch Speed Dome

DS-2TD4137T-9/W(B)	HM-TD4167-25/WY	HM-TD4137-25/WY
DS-2TD4137T-25/W(B)	HM-TD4167-50/WY	HM-TD4137-50/WY
DS-2TD4167T-9/W(B)	DS-2TD4167T-25/W(B)	□

2. 62x7 series: W Series Positioning System

DS-2TD6237-50H4L/W	DS-2TD6237-75C4L/W	DS-2TD6267-50H4L/W
DS-2TD6267-75C4L/WY	DS-2TD6267-100C4L/WY	DS-2TD6237T-25H4L/W
DS-2TD6237T-50H4L/W	DS-2TD6267T-25H4L/W	DS-2TD6267T-50H4L/W

3. TD63x8 series: T2Y Series Positioning System

HM-TD6338-50C4L/G0/T2Y	HM-TD6338-75C4L/G0/T2Y	HM-TD6368-50C4L/G0/T2Y
HM-TD6368-75C4L/G0/T2Y	HM-TD6368-100C4L/G0/T2Y	HM-TD6368-130ZC4L/G0/T2Y
HM-TD63C8-75C4L/G0/T2Y	HM-TD63C8-100C4L/G0/T2Y	□

4. TD81x7 series: W Series Stable PTZ

DS-2TD8167-150ZC4F/W	DS-2TD8167-150ZE2F/W(B)	DS-2TD8167-190ZE2F/W(B)
DS-2TD8167-190ZE2F/WY(B)	DS-2TD8167-230ZG2F/W	DS-2TD8167-230ZG2F/WY
DS-2TD8167-190ZE2FL/W(B)	DS-2TD8167-230ZG2FL/W	HM-TD8167-150ZC4F/WY
HM-TD8167-150ZE2F/WY(B)	HM-TD8167-190ZE2F/WY(B)	HM-TD8167-230ZK4F/WY
HM-TD8167-190ZE2FL/WY(B)	□	□

5. TD95C8 series: HD Sphere PTZ

HM-TD95C8-150ZE4FL/W(B)	HM-TD95C8-190ZK4FL/W(B)	HM-TD95C8-300ZK4FL/W(B)
DS-2TD95C8-150ZE4FL/W	DS-2TD95C8-190ZK4FL/W	DS-2TD95C8-300ZK4FL/W

6. TD966A series: Cooled Sphere PTZ

DS-2TD966A-450ZK2FD-W(B)	DS-2TD966A-750ZK2FD-W(B)	DS-2TD966A-1100ZK2FD-W(B)
DS-2TD966A-450ZK2F/W	DS-2TD966A-750ZK2F/W	DS-2TD966A-1100ZK2F/W

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

FIRMWARE

Users can download the latest baseline firmware from HIKVISION website:

<https://www.hikvision.com/en/support/download/firmware/?q=firmware&pageNum=1&position=1&hiksearch=true>

If you can't find firmware, please contact with technical support in your country.

WHAT IS LINKAGE TRACKING

1. Background

When you are using the radar or panoramic scanner together with thermal PTZ camera.

2. What does radar or scanner do?

Detect targets, send coordinates of targets to thermal PTZ camera.

3. What does thermal PTZ camera do?

Receive the coordinates, rotate to that direction, and then it can do two things:

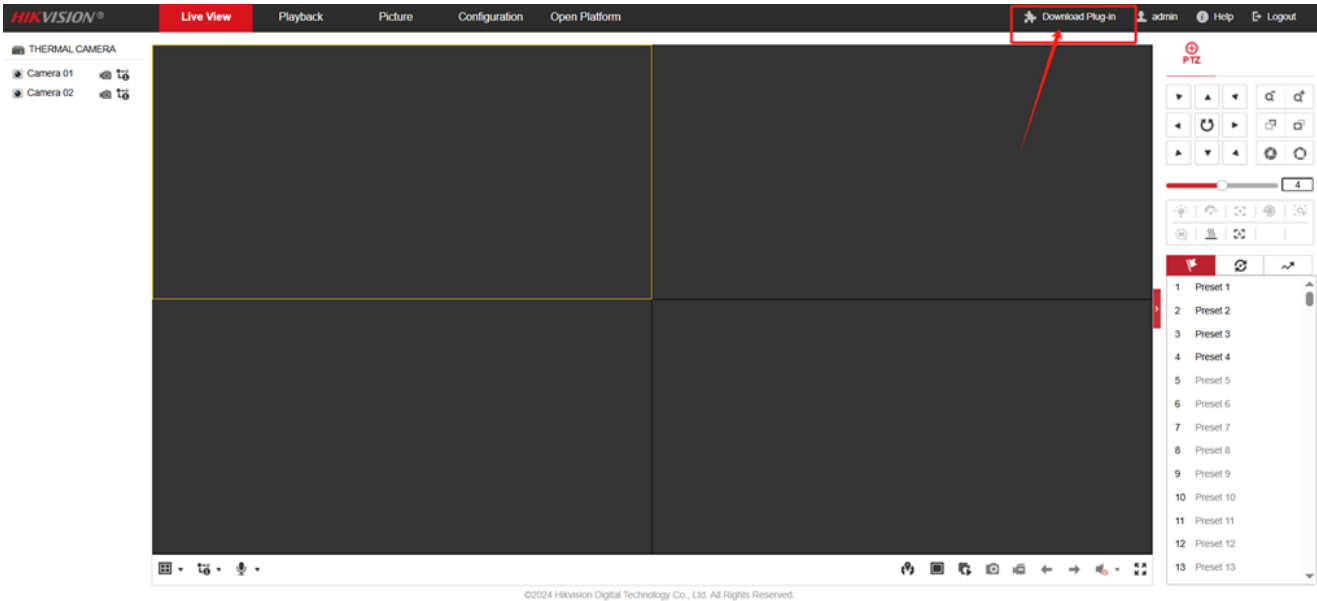
- PTZ camera will rotate and track the targets based on the constantly refreshing coordinates from radar.
- After Radar sends coordinates, PTZ camera will rotate to the position and try to distinguish the targets (vessel, vehicle, human), and then use the smart algorithm to track the target by camera itself.

HOW TO CONFIGURE LINKAGE TRACKING

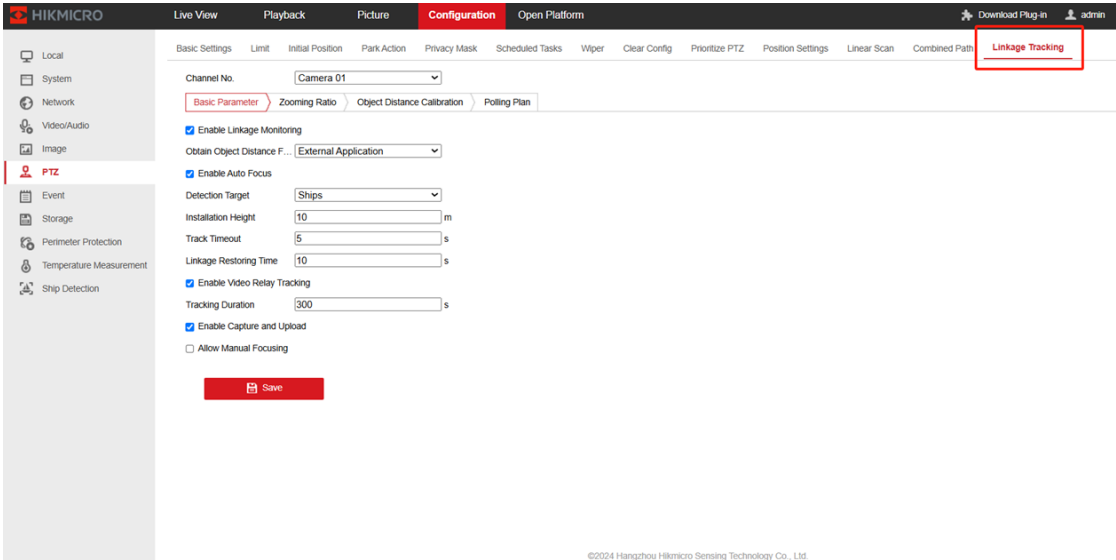
1. Preparation

- Please make sure you already finished the radar and thermal camera direction calibration. This document is only for thermal PTZ camera side configuration.
- Please check the firmware of thermal PTZ camera. We recommend to download the latest firmware from our official website: <https://www.hikvision.com/en/>
- Login to camera's IP, download and install the plug-in. Please close the webpage, install the plugin, then open the webpage and re-login the camera.

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		



○ Go to Configuration—PTZ—Linkage Tracking



2. Polling Plan

Camera supports optical channel (camera01) or thermal channel (camera02) for linkage tracking. Please make sure you configure the right channel.

For example, in one project, users need optical channel for linkage tracking, so from now on, all below configuration is based on camera01.

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

Channel No.

Camera 01

Basic Parameter

Zooming Ratio

Object Distance Calibration

Polling Plan

Polling Mode

Manual

Optical Channel

✕ Delete

🗑️ Delete All

Mon

024681012141618202224

Tue

024681012141618202224

Wed

024681012141618202224

Thu

024681012141618202224

Fri

024681012141618202224

Sat

024681012141618202224

Sun

024681012141618202224

Optical Channel

Thermal Channel

Save

3. Basic Parameter

- Track Timeout: Input the stopping time of tracking. The camera will stop tracking if there is no updated position from the radar beyond this time.
- Linkage Restoring Time: Input the restoring time of tracking. The camera will restore to linkage tracking after this time if the tacking process has been interrupted by PTZ manual control or presets calling.
- Enable Video Relay Tracking. When the specific target is detected by the camera repeatedly, the radar tracking system will switch to the video tracking. The camera will continuously track the target based on the video detection in Tracking Duration.

4 / 9

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

HIKMICRO

Live View

Playback

Picture

Configuration

Open Platform

Local

System

Network

Video/Audio

Image

PTZ

Event

Storage

Perimeter Protection

Temperature Measurement

Ship Detection

Basic Settings

Limit

Initial Position

Park Action

Privacy Mask

Scheduled Tasks

Wiper

Channel No.

Camera 01

Basic Parameter

Zooming Ratio

Object Distance Calibration

Polling Plan

☒ Enable Linkage Monitoring

Obtain Object Distance F...

External Application

☒ Enable Auto Focus

Detection Target

Ships

Installation Height

10

 m

Track Timeout

5

 s

Linkage Restoring Time

10

 s

☒ Enable Video Relay Tracking

Tracking Duration

300

 s

☒ Enable Capture and Upload

☐ Allow Manual Focusing

Save

4. Zooming Ratio

Set the automatic zooming ratio in different object distance ranges to view the moving target in proper image size during the tracking process.

Please firstly choose the right channel no.

You can use Get Recommended Value, or Manual Add.

5 / 9

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

Detection Info. ✕

Target Size

Length m ✓

Width m ✓

Detection Range

Min m ✓

Max m

Target to Image Ratio (%)

Min % ✓

Max % ✓

Tracking Accuracy

Tracking Accuracy Degree ✓

OK

Cancel

Zooming Ratio Get Recommended Value Manual Add Delete

No.	Lower Limit of Object Distance(m)	Upper Limit of Object Distance(m)	Rate
1	100	174	16.6
2	174	300	19.5

Save

4.2 Manual Add

Or you can manually add several zooming ratios.

Zooming Ratio Get Recommended Value Manual Add Delete

No.	Lower Limit of Object Distance(m)	Upper Limit of Object Distance(m)	Rate
1	50	200	15
2	200	400	20
3	400	550	23

Save

5. Object Distance Calibration

Set and calibrate the automatic zooming ratio and focusing value in specific object distance to get better image quality in scenes of darkness and low temperature difference.

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

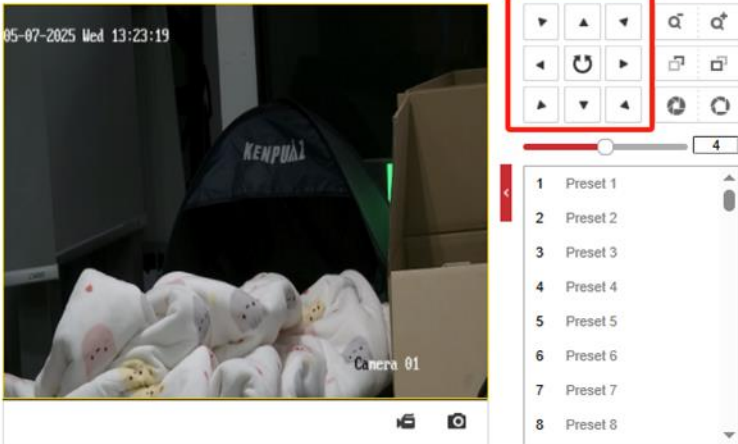
Channel No. Camera 01

Basic Parameter Zooming Ratio **Object Distance Calibration** Polling Plan

⚠ Please configure the Tracking Zoom Ratio first.

Object Distance Calibra...

1. Aim the device at the target that meets the distance requirements. Manually adjust the zoom ratio to the maximum, or keep th...



2. Enter the actual value of the object distance, and click Search and then Calibration buttons to complete a set of calibration.

Object Distance Calibra...

Actual Object Distance(m)	Temperature(Degre...	Rate	Focus	Operation		
3	41	11	43887	Search	Calibration	View Location
1	41	3	35571	Search	Calibration	View Location
2	41	7	42982	Search	Calibration	View Location

- Choose the channel (camera01 or camera02)
- Rotate camera, search for the real target, and aim the device at the target that meets the distance requirements. Adjust the zooming ratio to the maximum with the PTZ control panel, and adjust the focus until the target is clear enough.
- Type in the Actual Object Distance, then click “search”, then click “calibration”. One object distance calibration is finished.
- Optical channel needs at least 3 objects, thermal channel needs at least 1 object.

6. Final Checking

- Obtain Object Distance from External Application or PTZ Calculation
- External Application: Obtain the distance from Radar
- PTZ Calculation: Obtain by camera’s algorithm

Product	Thermal PTZ Cameras	Version	V1.0
Contact	lixiaowen7@hikmicrotech.com	Date:	2025/6/5
Title	How to Configure Linkage Tracking for Thermal PTZ Cameras		

- If you find during the linkage tracking, the image is not clear enough, you may enable auto focus.

Channel No. Camera 01

Basic Parameter Zooming Ratio Object Distance Calibration Polling Plan

☒ Enable Linkage Monitoring

Obtain Object Distance Fr. External Application

☐ Enable Auto Focus

Detection Target

Installation Height 10 m

Track Timeout 5 s

Linkage Restoring Time 10 s

☒ Enable Video Relay Tracking

Tracking Duration 300 s

☒ Enable Capture and Upload

☐ Allow Manual Focusing

 Save